

Application Layer

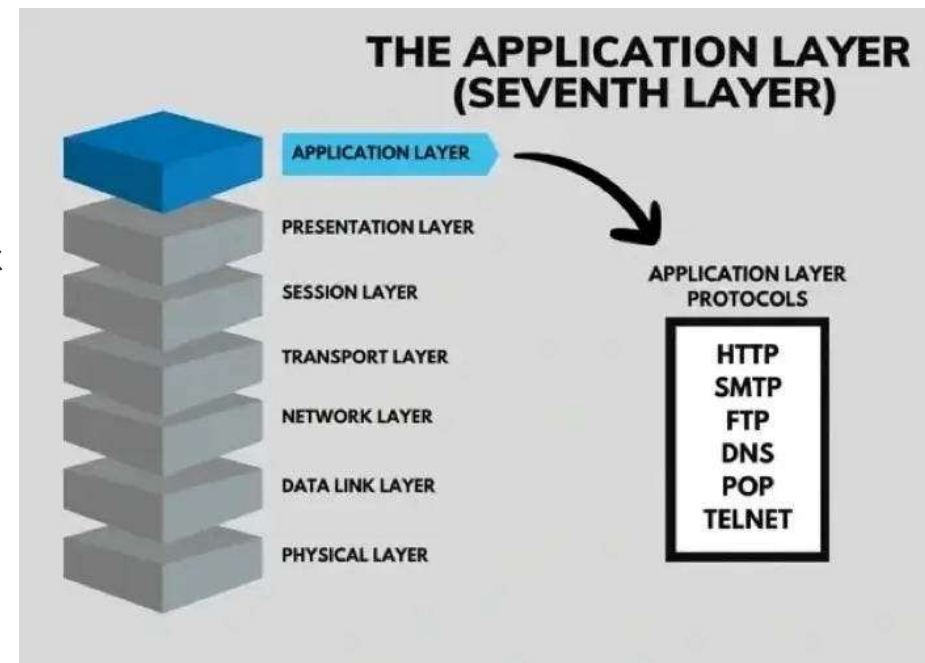


School of Computer Science Engineering and Technology
Bennett University
Greater Noida – 201310

Application Layer

The Application Layer is the topmost layer (Layer 7) of the OSI Model that enables direct interaction between end-user applications and the network.

Closest layer to the end user
Provides network services to applications
Enables file transfer, email, web access, and remote login
Acts as an interface between user software and the network



Services Provided by Application Layer Protocols

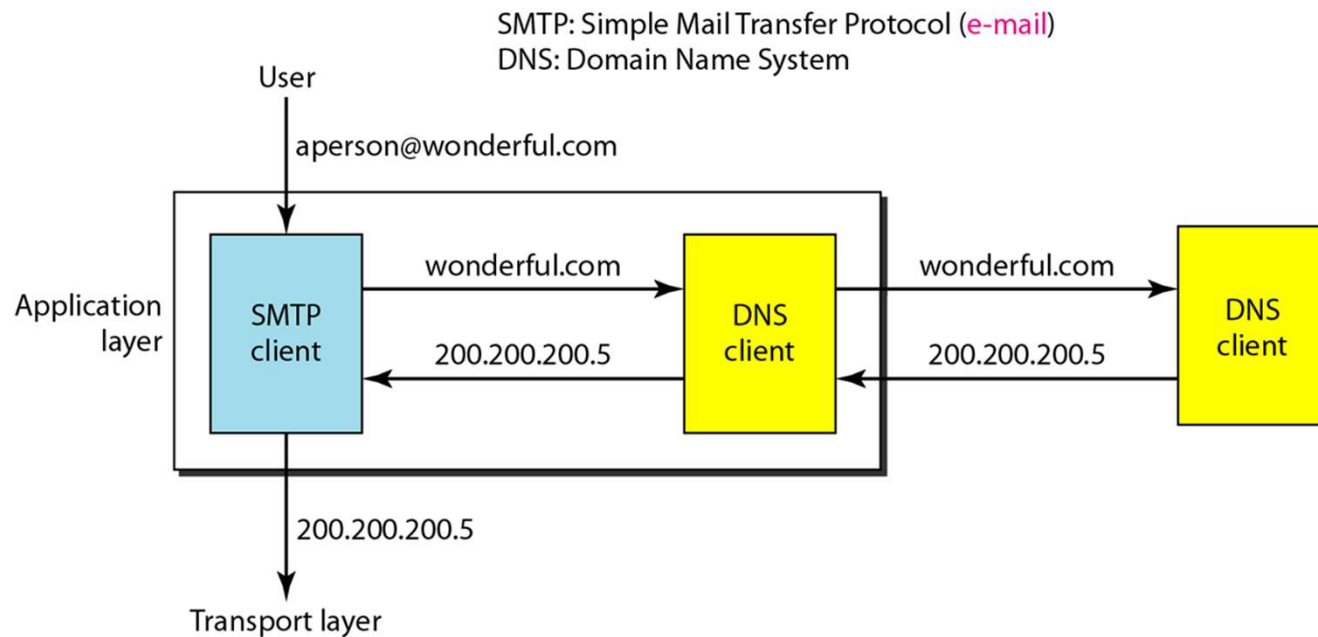
- The Application Layer protocol defines process for both parties which are involved in communication.
- These protocols define the type of message being sent or received from any side (either source host or destination host).
- These protocols also define basic syntax of the message being forwarded or retrieved.
- These protocols define the way to send a message and the expected response.
- These protocols also define interaction with the next level.

Protocols of the Application Layer

The application layer provides several protocols which allow any software to easily send and receive information and present meaningful data to its users. The following are some of the application layer protocols.

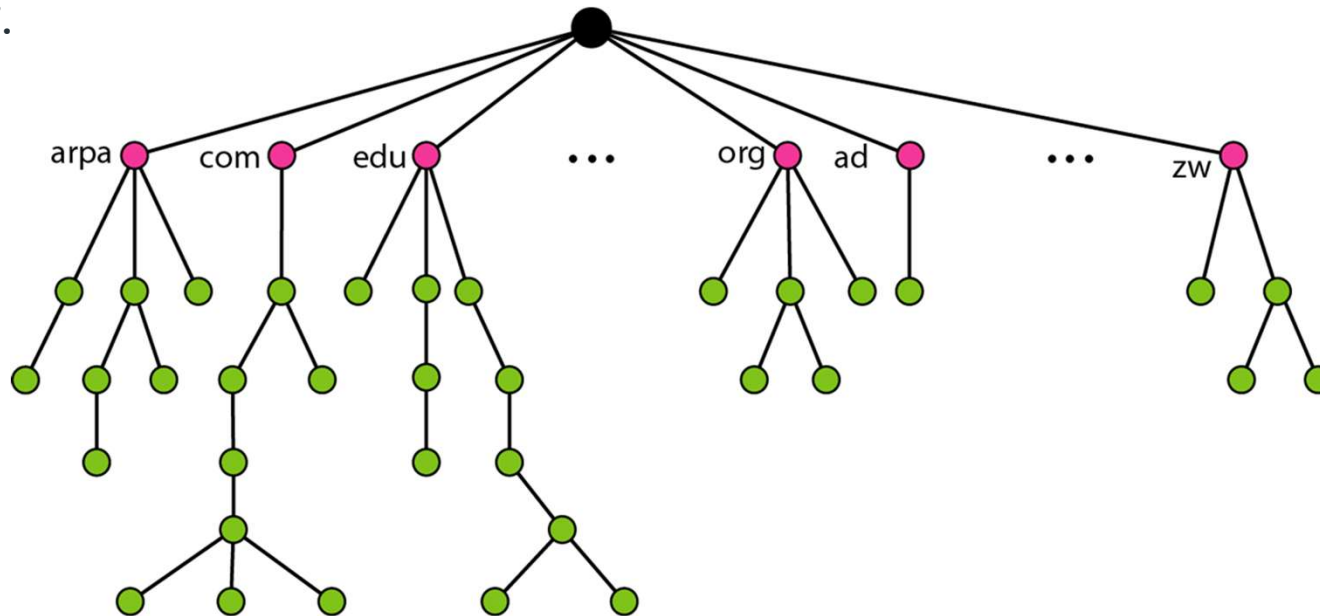
- **HTTP (HyperText Transfer Protocol):** Used for web communication (Port 80).
- **DNS (Domain Name System):** Translates domain names into IP addresses (Port 53).
- **TELNET:** Provides remote login and file management services (Port 23).
- **DHCP (Dynamic Host Configuration Protocol):** Assigns IP addresses to hosts dynamically (Ports 67 & 68).
- **FTP (File Transfer Protocol):** Facilitates file transfer between systems (Ports 20 for data, 21 for control).
- **SMTP (Simple Mail Transfer Protocol):** Handles sending of emails (Ports 25 & 587).
- **NFS (Network File System):** Enables remote access to files as if they were local (Port 2049).
- **SNMP (Simple Network Management Protocol):** Manages and monitors devices in a network (Ports 161 & 162).

Domain Name System, DNS

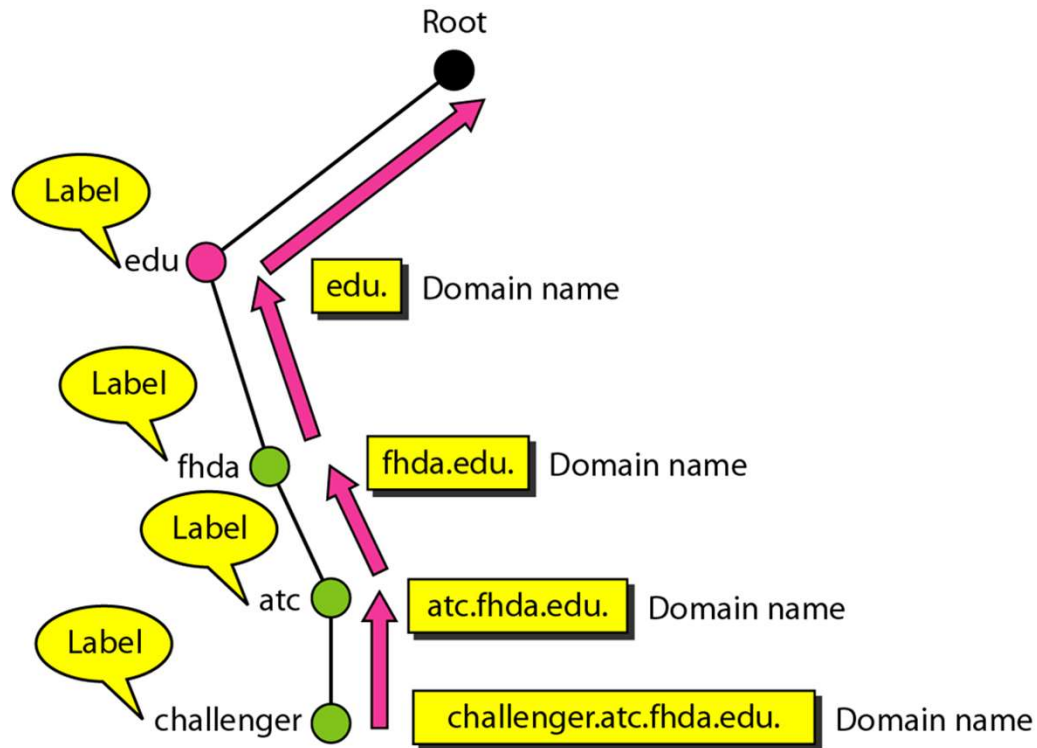


Domain Name Space

To have a hierarchical name space, a domain name space was designed. In this design the names are defined in an inverted-tree structure with the root at the top. The tree can have only 128 levels: level 0 (root) to level 127.



Domain names and labels



FQDN and PQDN

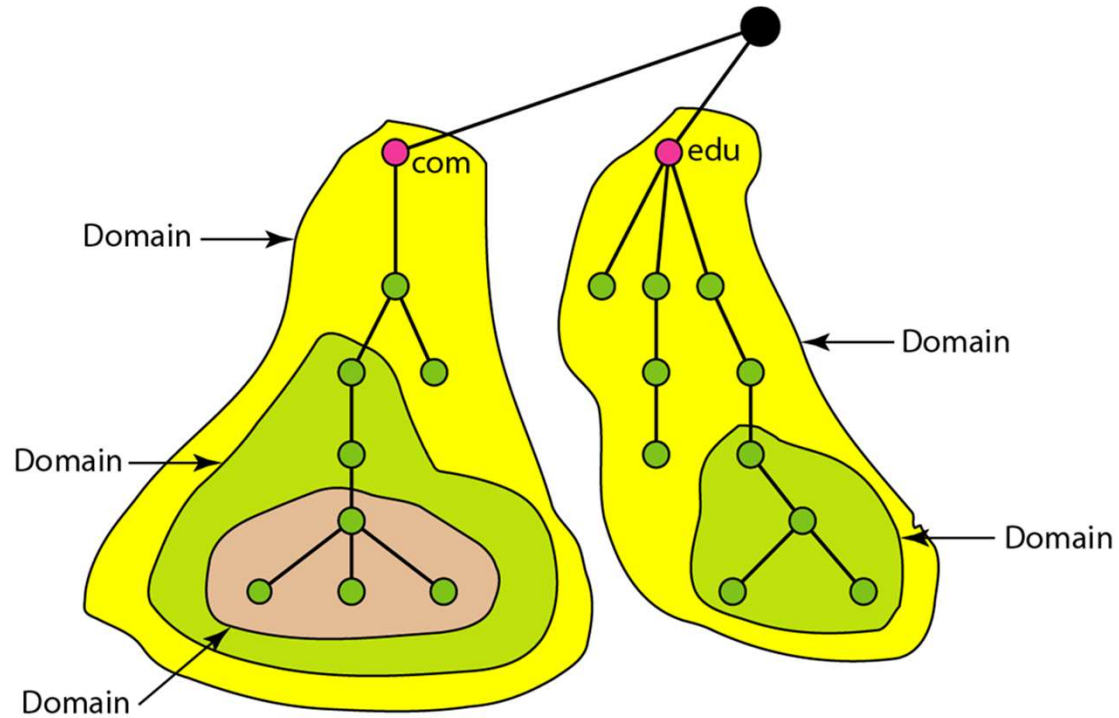
FQDN

challenger.atc.fhda.edu.
cs.hmme.com.
www.funny.int.

PQDN

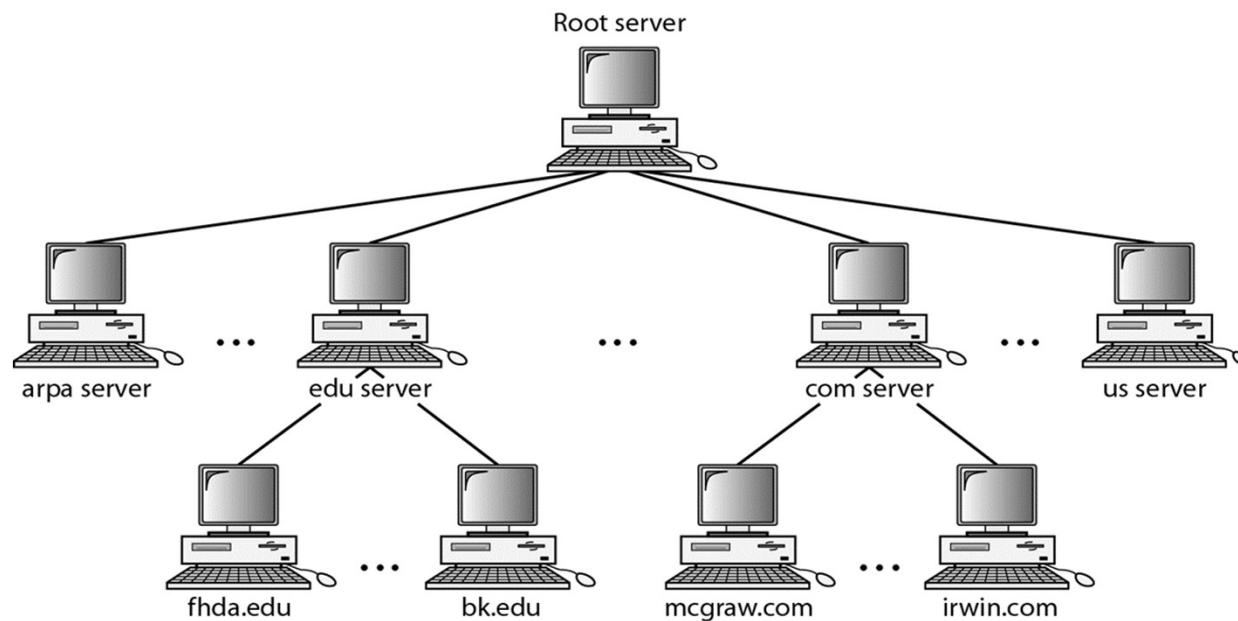
challenger.atc.fhda.edu
cs.hmme
www

Domains

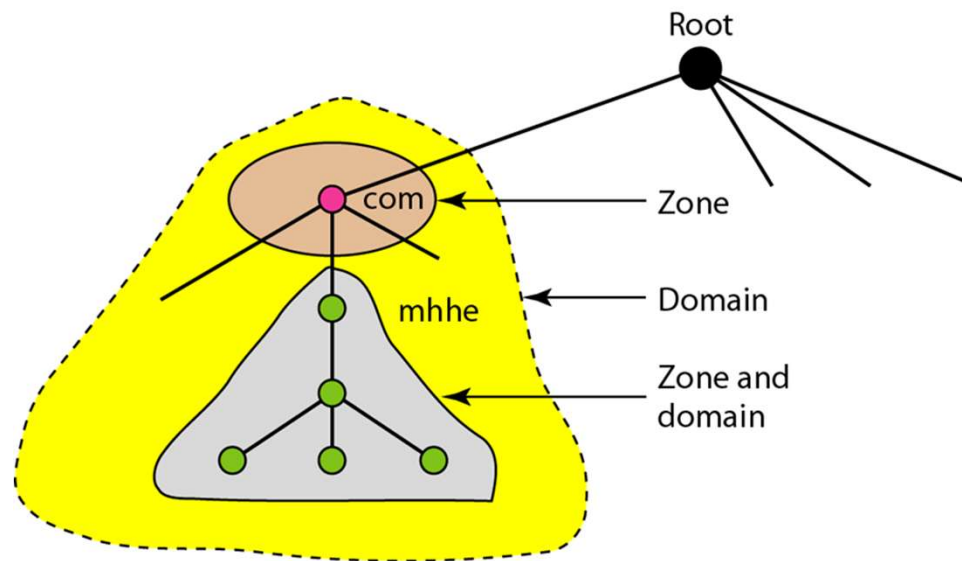


DISTRIBUTION OF NAME SPACE

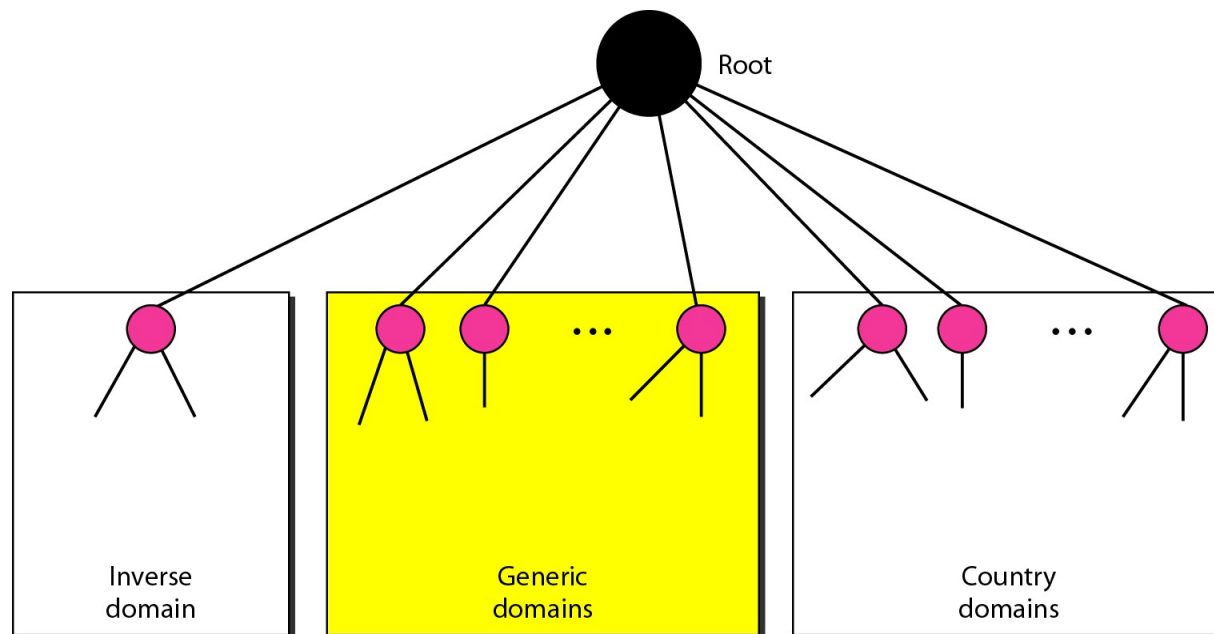
The information contained in the domain name space must be stored. However, it is very inefficient and also unreliable to have just one computer store such a huge amount of information. In this section, we discuss the distribution of the domain name space.



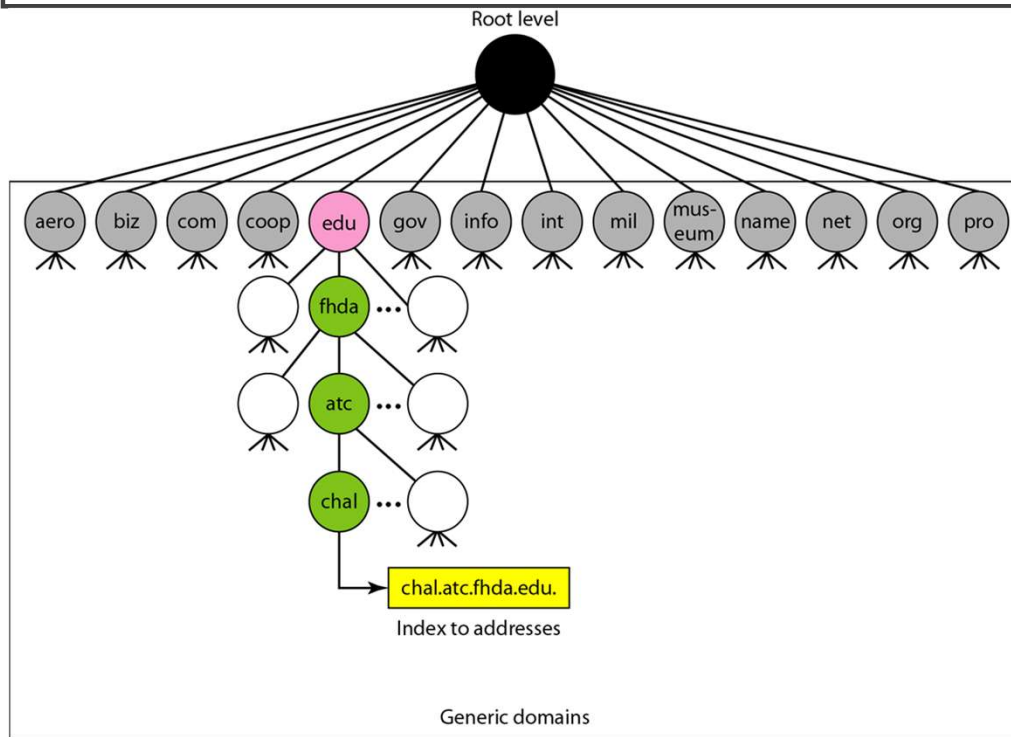
Zones and domains



DNS in Internet

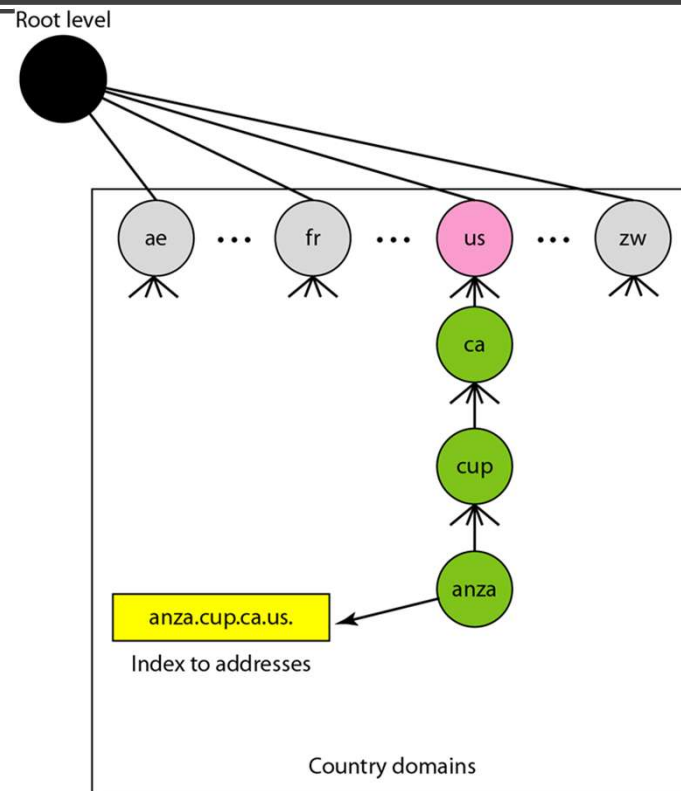


Generic Domains

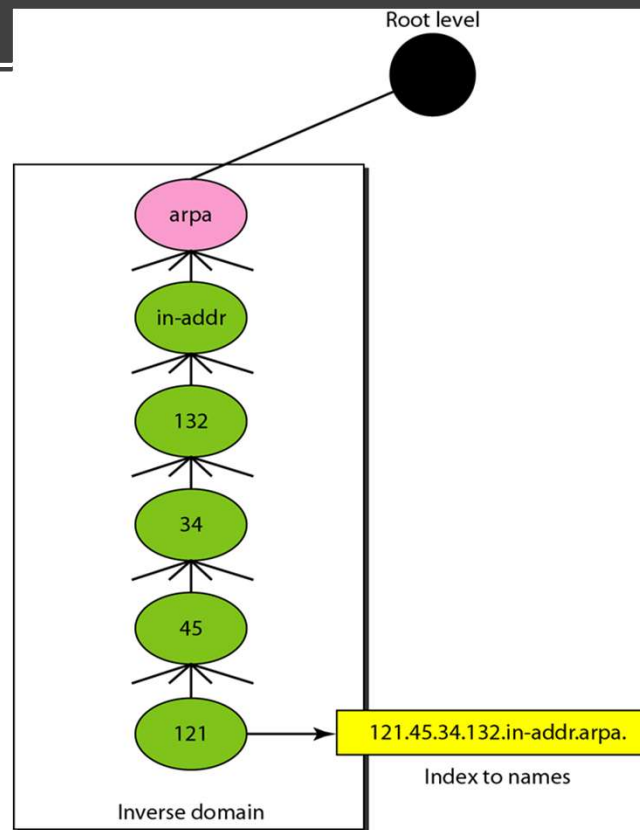


<i>Label</i>	<i>Description</i>
aero	Airlines and aerospace companies
biz	Businesses or firms (similar to “com”)
com	Commercial organizations
coop	Cooperative business organizations
edu	Educational institutions
gov	Government institutions
info	Information service providers
int	International organizations
mil	Military groups
museum	Museums and other nonprofit organizations
name	Personal names (individuals)
net	Network support centers
org	Nonprofit organizations
pro	Professional individual organizations

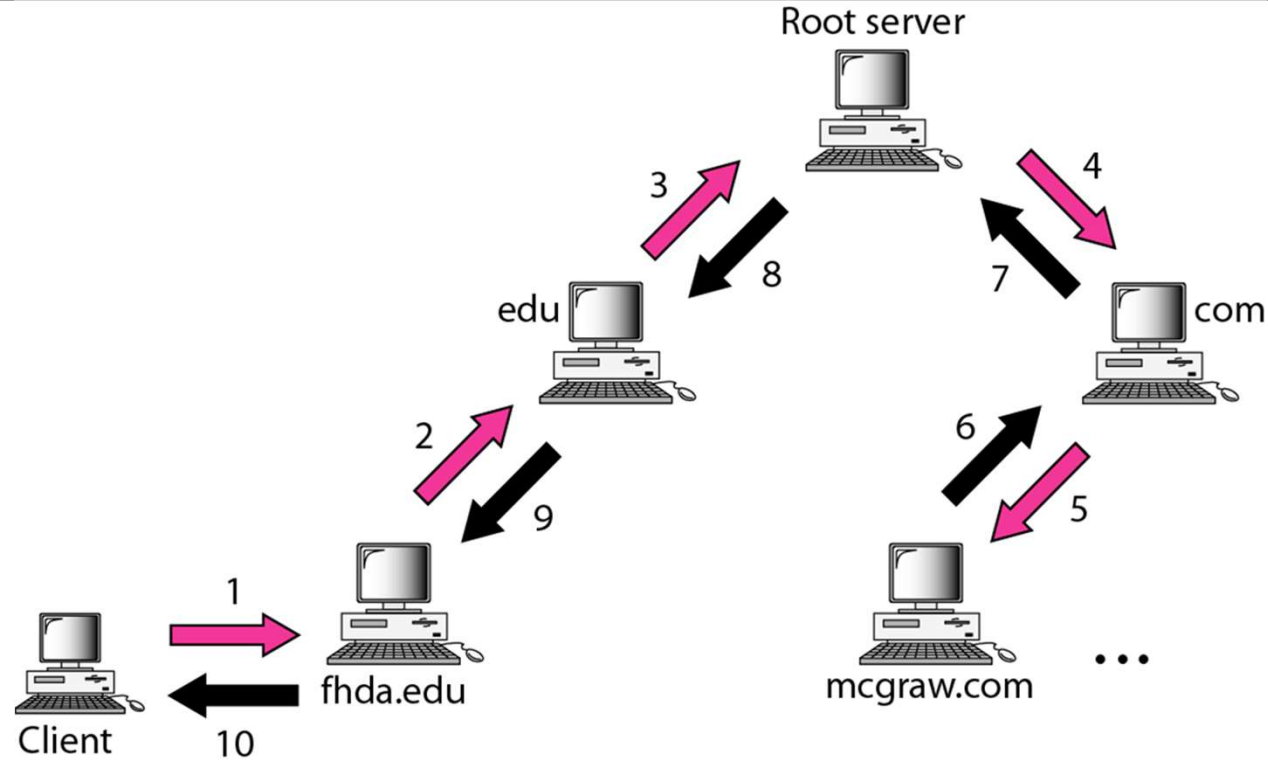
Country Domains



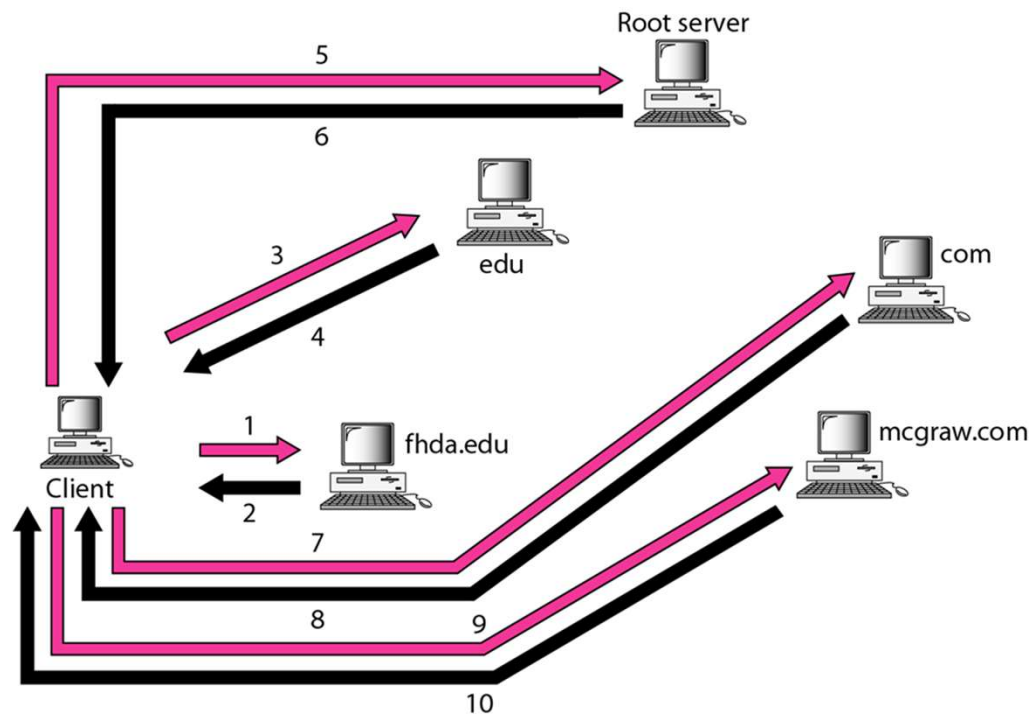
Inverse domain



RESOLUTION: Recursive

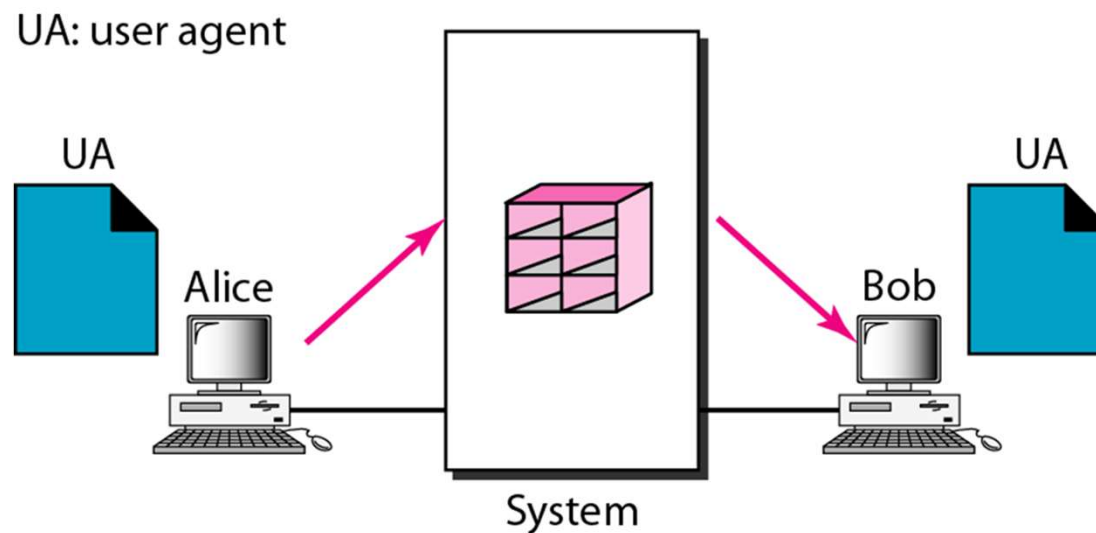


RESOLUTION: Iterative



ELECTRONIC MAIL: The 4 Push/Pull Scenarios

First scenario in electronic mail

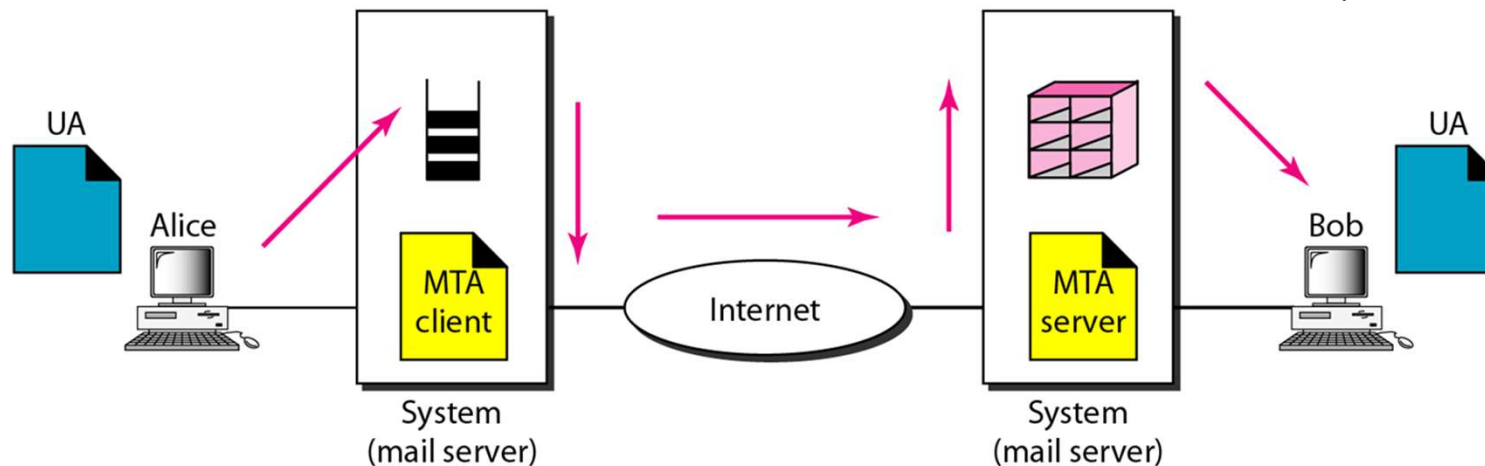


When the sender and the receiver of an e-mail are on the same system, we need only two user agents.

ELECTRONIC MAIL: The 4 Push/Pull Scenarios

Second scenario in electronic mail

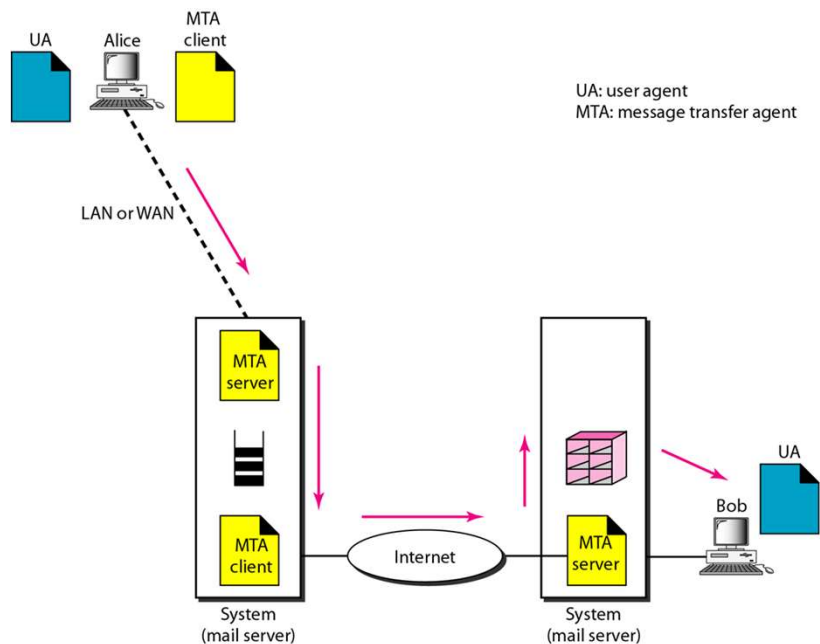
UA: user agent
MTA: message transfer agent



When the sender and the receiver of an e-mail are on different systems, we need two UAs and a pair of MTAs (client and server).

ELECTRONIC MAIL: The 4 Push/Pull Scenarios

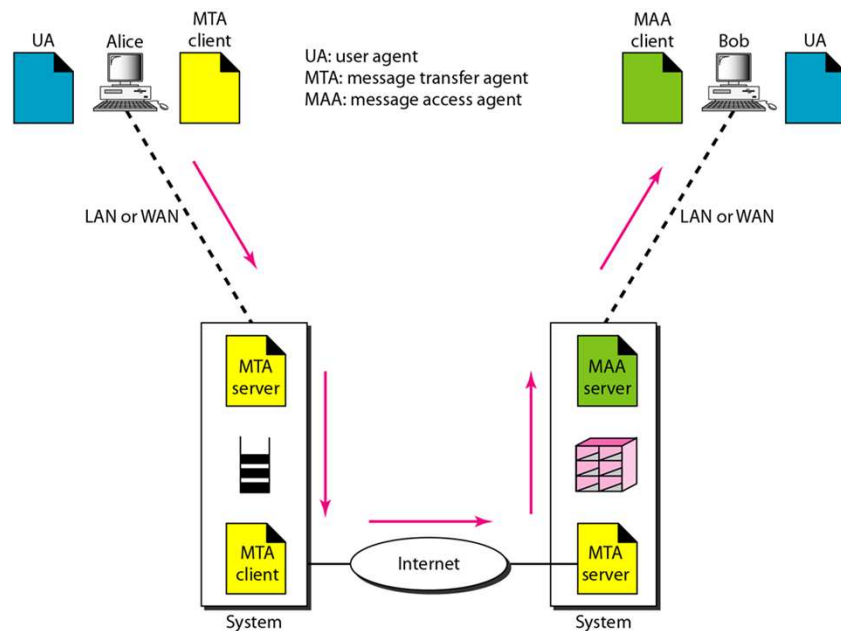
Third scenario in electronic mail



When the sender is connected to the mail server via a LAN or a WAN, we need two UAs and two pairs of MTAs (client and server).

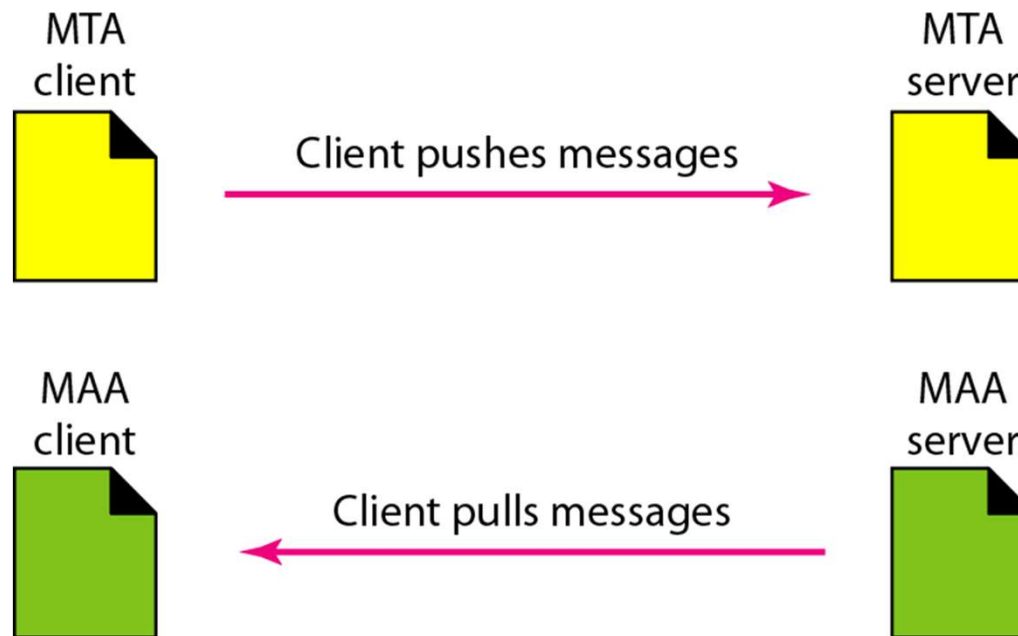
ELECTRONIC MAIL: The 4 Push/Pull Scenarios

Fourth scenario in electronic mail

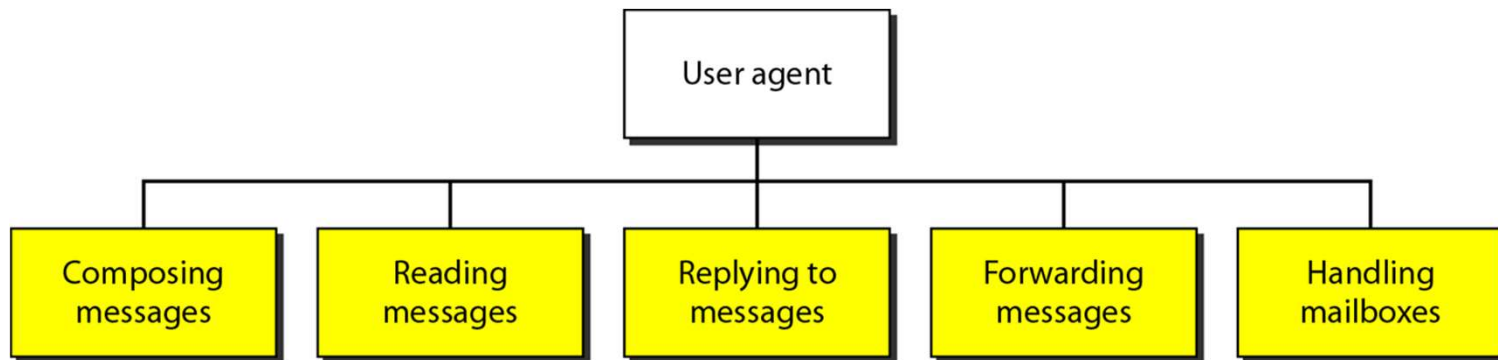


When both sender and receiver are connected to the mail server via a LAN or a WAN, we need two UAs, two pairs of MTAs and a pair of MAAs.

Push versus pull in electronic email



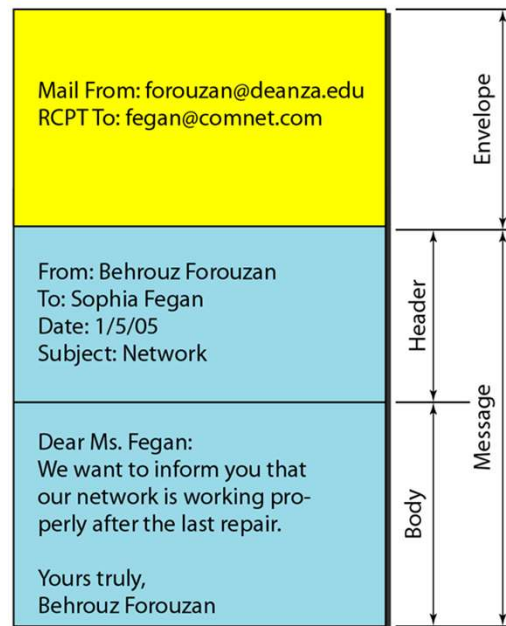
Services of user agent



Services of user agent



a. Postal mail

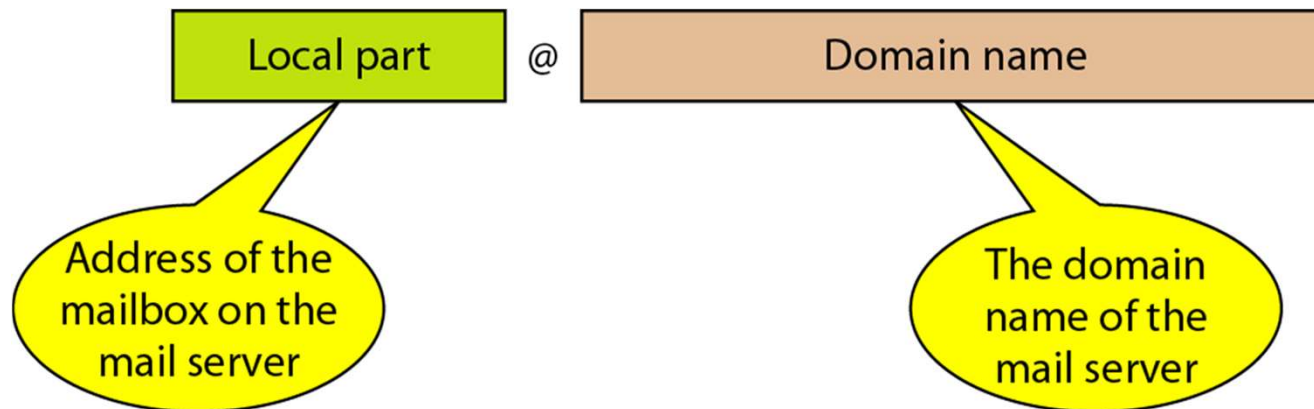


b. Electronic mail

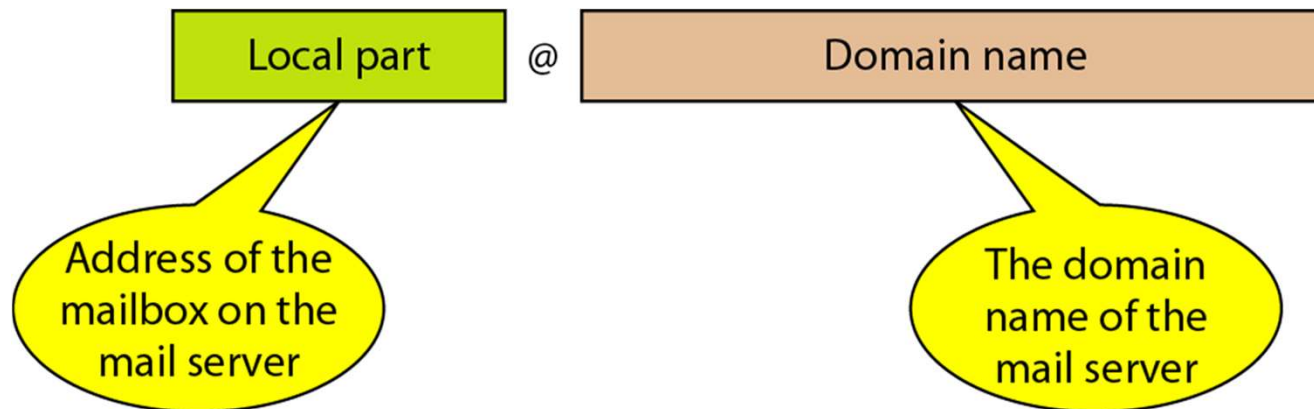
Some examples of command-driven user agents are mail, pine, and elm.

Some examples of GUI-based user agents are Eudora, Outlook, and Netscape.

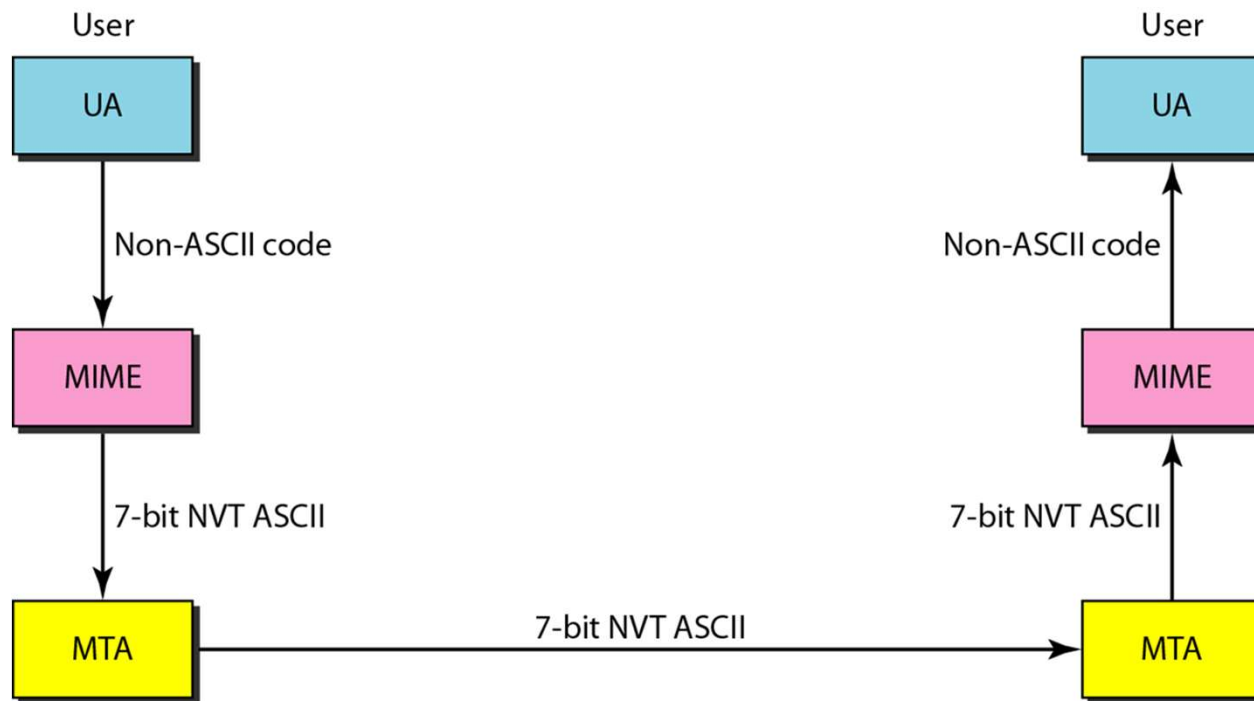
E-mail address



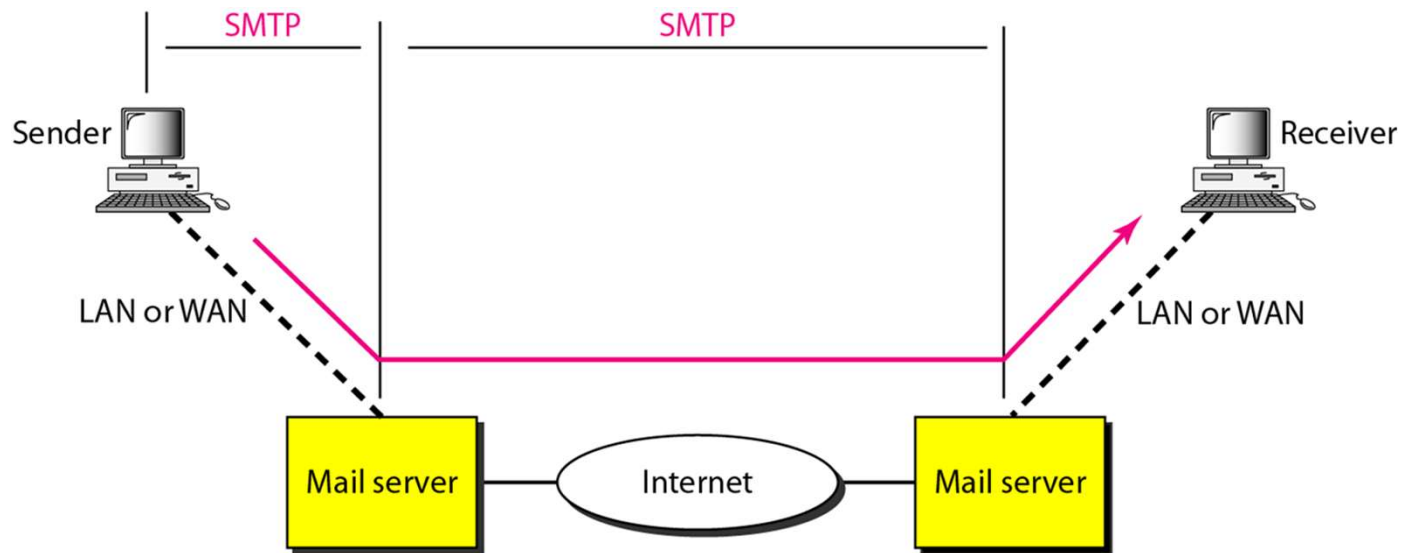
E-mail address



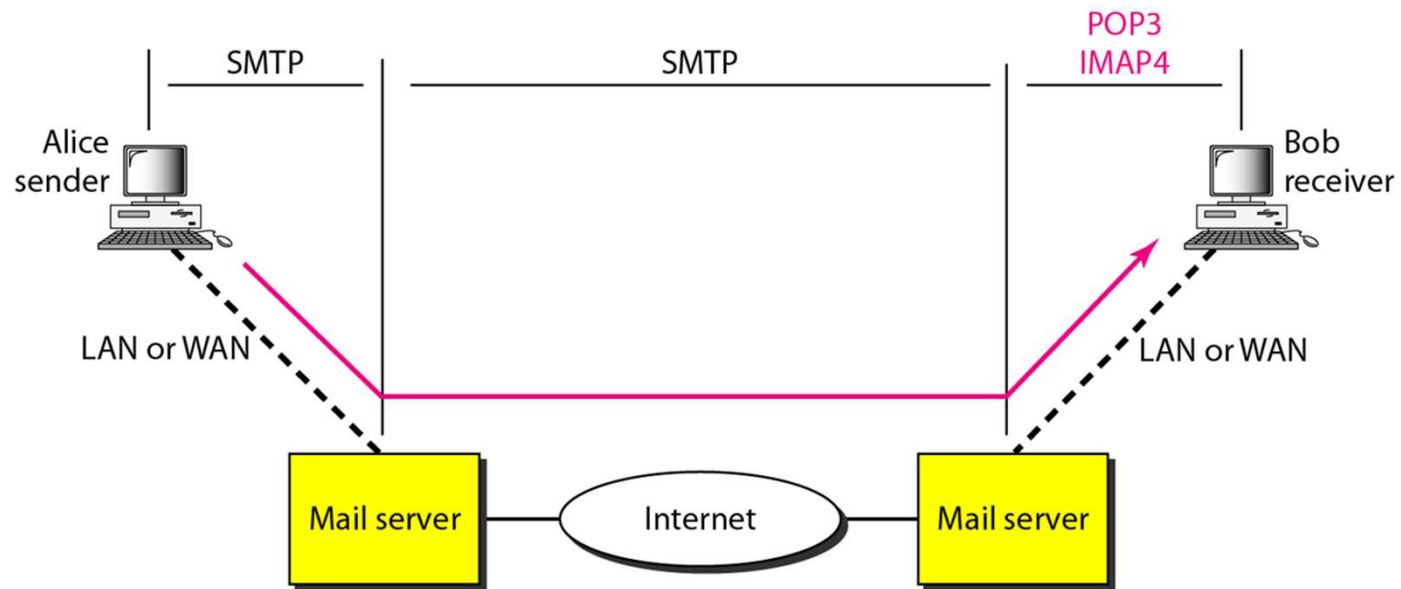
MIME



SMTP Range



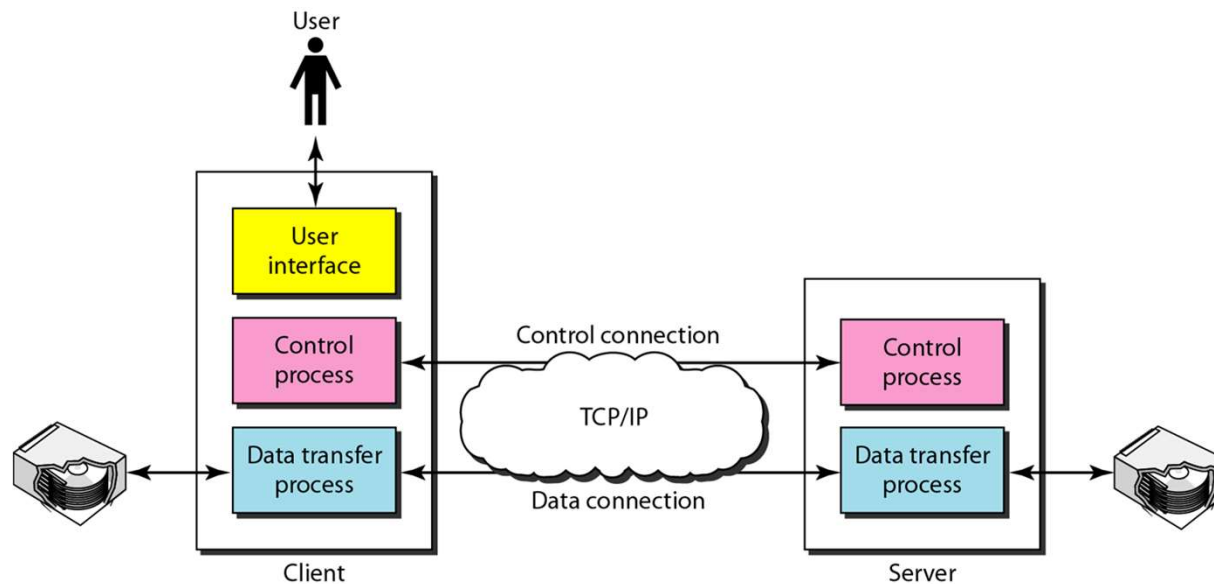
POP3 and IMAP4



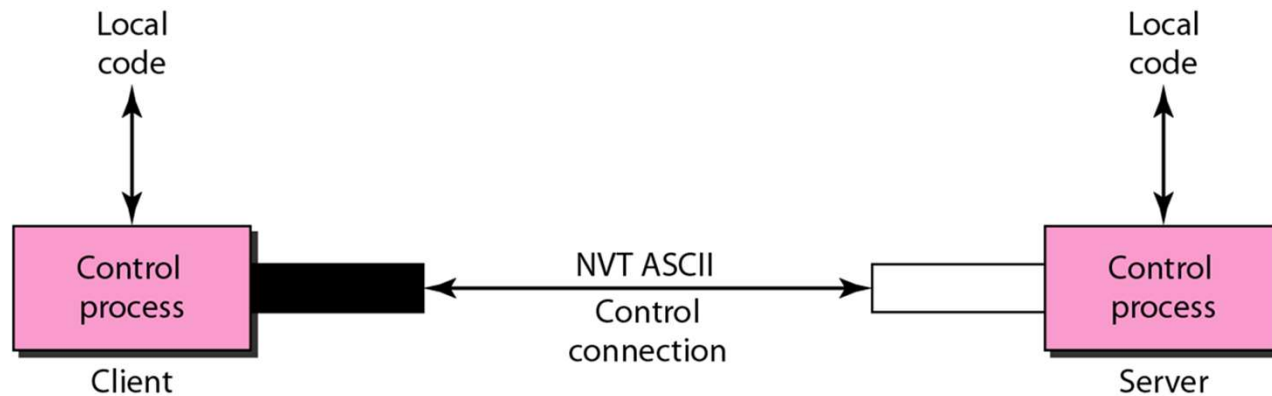
FTP

FTP uses the services of TCP. It needs two TCP connections.

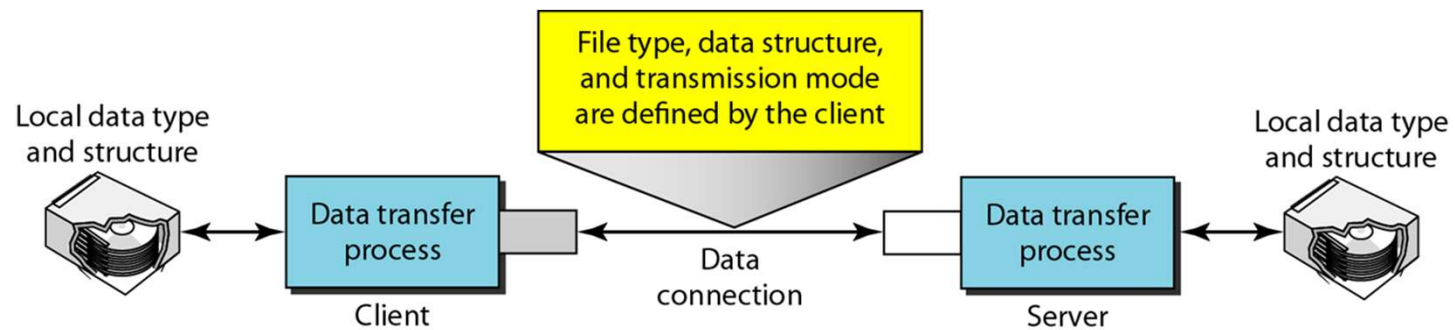
The well-known port 21 is used for the control connection and the well-known port 20 for the data connection



Using the control connection

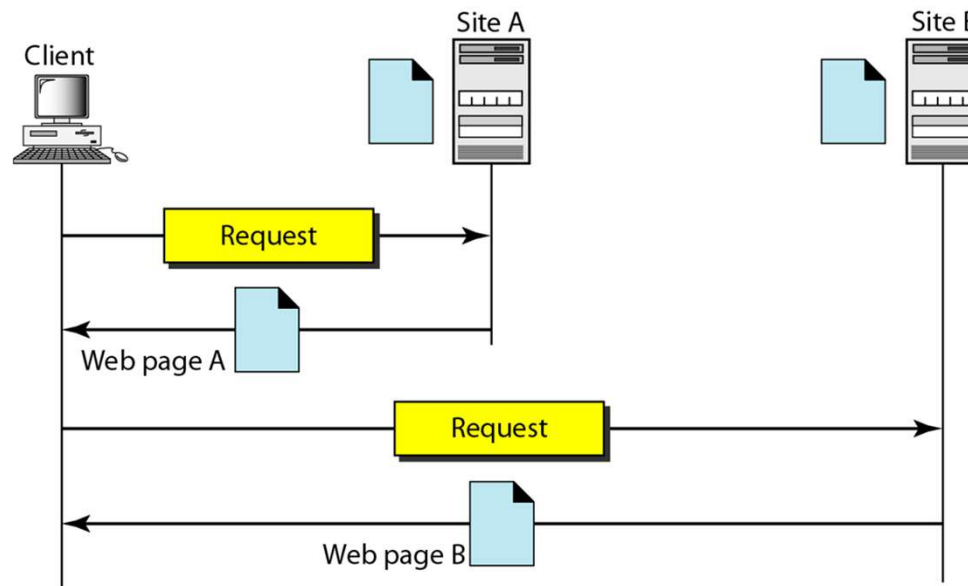


Using the data connection

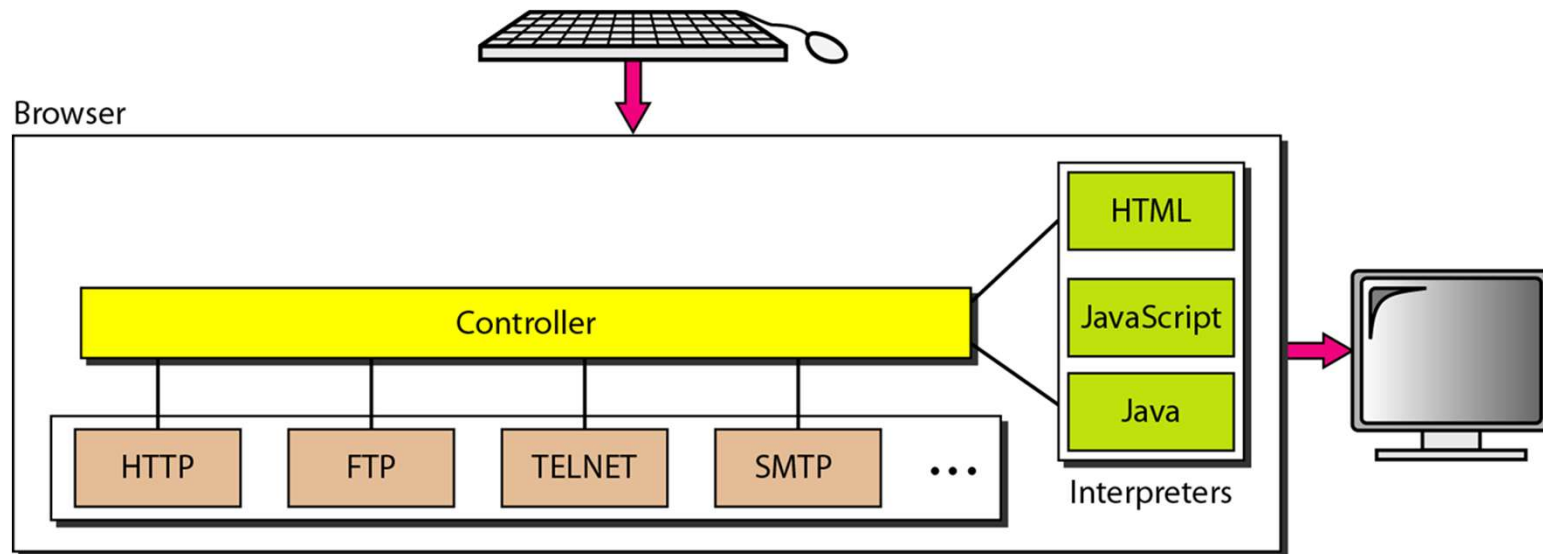


WWW

The WWW today is a distributed client/server service, in which a client using a browser can access a service using a server. However, the service provided is distributed over many locations called sites.



Browser



URL

Protocol

://

Host

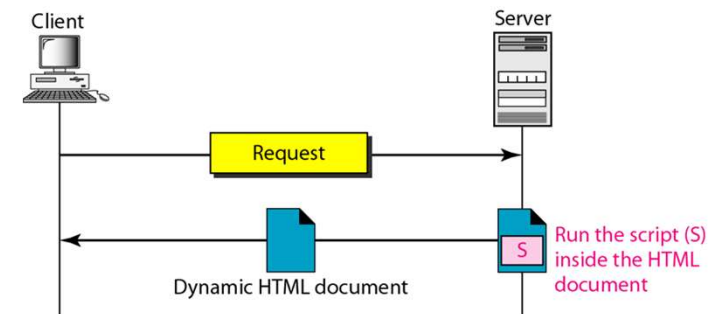
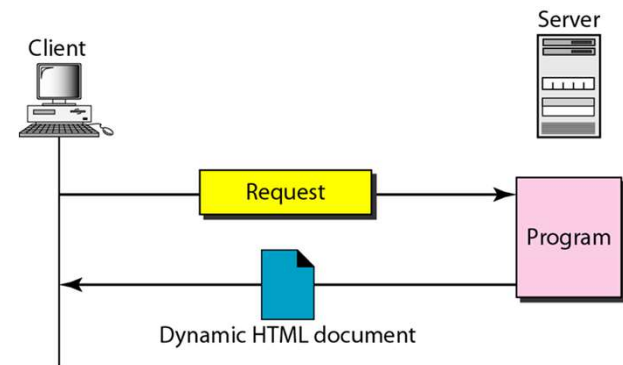
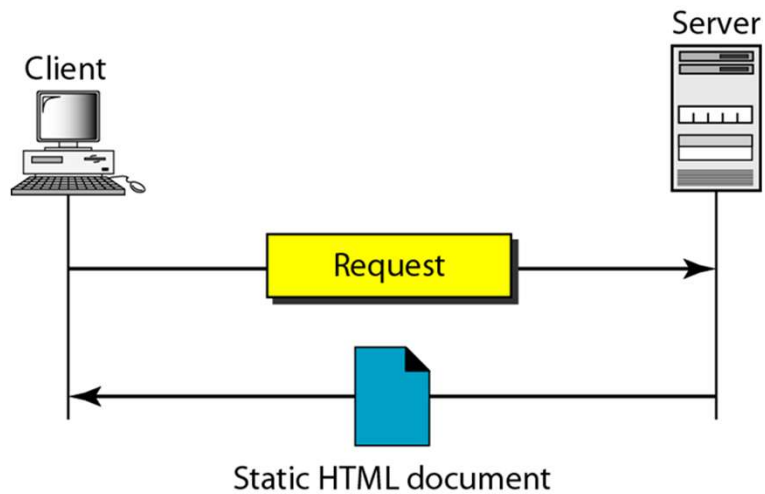
:

Port

/

Path

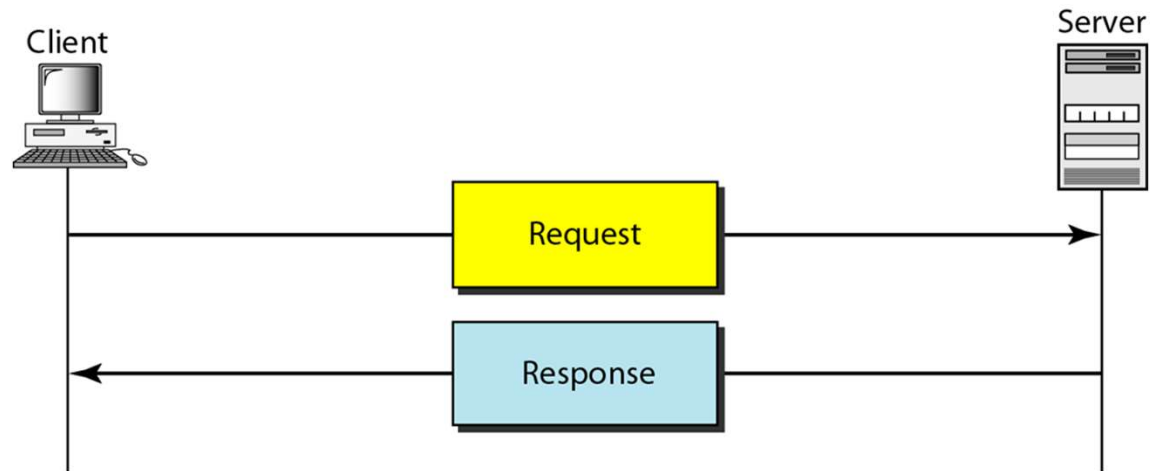
Static and Dynamic Documents



HTTP

The Hypertext Transfer Protocol (HTTP) is a protocol used mainly to access data on the World Wide Web. HTTP functions as a combination of FTP and SMTP.

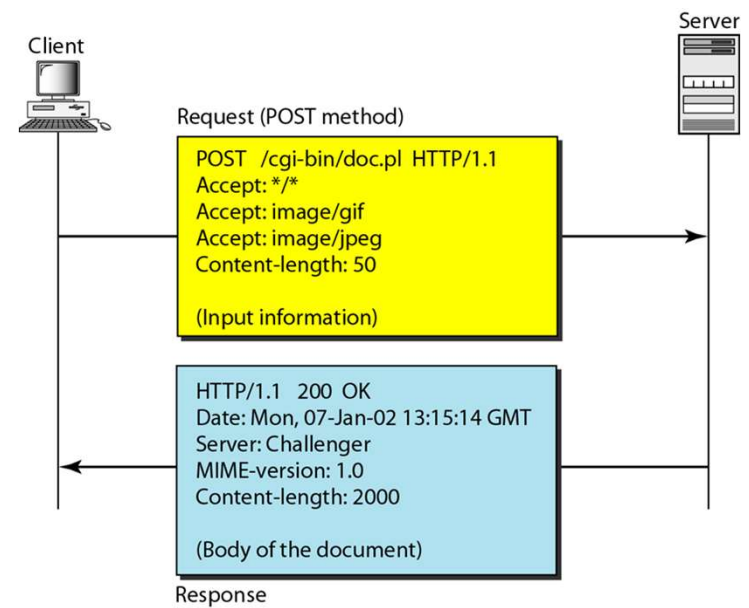
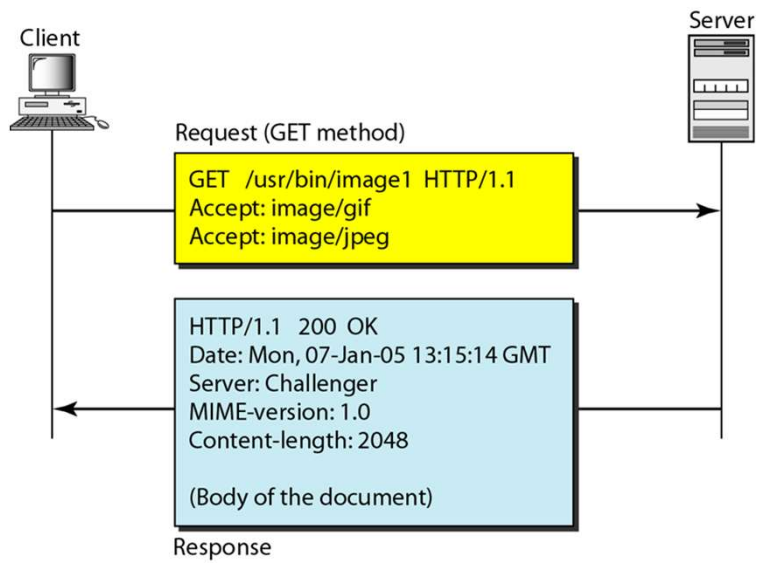
HTTP uses the services of TCP on well-known port 80.



HTTP Methods

<i>Method</i>	<i>Action</i>
GET	Requests a document from the server
HEAD	Requests information about a document but not the document itself
POST	Sends some information from the client to the server
PUT	Sends a document from the server to the client
TRACE	Echoes the incoming request
CONNECT	Reserved
OPTION	Inquires about available options

HTTP Example



Thank You
